BAYES’ THEOREM

**Module 1: The Illusion of Certainty **

1. CONDITIONAL PROBABILITY

Before we can use Bayes’ Theorem, we must understand how the universe changes when new information is revealed. Conditional probability is the probability of an event occurring *given that another event has already occurred*.

- **The Notation:** $P(A|B)$ is read as “The probability of A, given B.”
- **The Formula:** The probability of A given B is the probability of both happening together, divided by the probability of B happening at all.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

2. THE BASE RATE FALLACY

Human intuition is terrible at medical tests because we ignore the **Base Rate** (how common a condition is in the general population).

If a disease is extremely rare, even a highly accurate test will produce thousands of “False Positives.” The accuracy of the test alone does not tell you if you are sick; you must factor in the rarity of the disease.

3. BAYES’ THEOREM: THE FORMULA

Bayes’ Theorem allows us to mathematically “change our minds.” It takes our initial belief about the universe and explicitly updates it using new evidence.

The Standard Equation:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

The Anatomy of the Equation:

- $P(A)$ **| The Prior:** Your initial belief (or the baseline probability) before seeing any new evidence.
- $P(B|A)$ **| The Likelihood:** The probability of seeing this new evidence if your initial belief was actually true.
- $P(B)$ **| The Evidence (Marginal Likelihood):** The total probability of seeing this evidence under *any* circumstance (true or false).
- $P(A|B)$ **| The Posterior:** Your newly updated belief after seeing the evidence.

4. THE EXPANDED DENOMINATOR (LAW OF TOTAL PROBABILITY)

In the real world, calculating the total evidence $P(B)$ can be tricky. We usually have to split it into two realities: when A is true, and when A is false ($\neg A$).

We expand the bottom of the equation like this:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B|A) \cdot P(A) + P(B|\neg A) \cdot P(\neg A)}$$

5. THE MEDICAL TEST PROOF

Let's prove why a 90% accurate test is lying to you.

- Only **1%** of the population has a rare disease. Prior: $P(Disease) = 0.01$
- The test correctly identifies **90%** of sick people. True Positive: $P(+|Disease) = 0.90$
- The test incorrectly flags **9%** of healthy people. False Positive: $P(+|NoDisease) = 0.09$

If you test positive ($+$), what is the actual mathematical probability that you are sick? Let's plug it into the expanded equation:

$$P(Disease|+) = \frac{0.90 \cdot 0.01}{(0.90 \cdot 0.01) + (0.09 \cdot 0.99)}$$

$$P(Disease|+) = \frac{0.009}{0.009 + 0.0891}$$

$$P(Disease|+) = \frac{0.009}{0.0981} \approx 0.0917$$

The Shocking Truth: Even with a positive result from a highly accurate test, you only have a **9.17%** chance of actually having the disease. The Base Rate overrides the test.